Department of Computer Engineering

CSE5041 Database Design & Development
Project Report

PROJECT TITLE

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1 INTRODUCTION

1.1 PROJECT DESCRIPTION

The Family Budget Database stores information about family members, financial accounts, transactions, and budgeting goals. The following data requirements have been identified and represented in the design:

- **User Management:** The database stores each Family Member's personal details, including name, email (for login) and their role within the family (e.g., Parent or Child). This distinction allows for future implementation of permission levels.

- **Account Management:** The system tracks various financial Accounts (e.g., Checking, Savings, Credit Cards, Cash Wallets). Each account is defined by a type and a current balance. The database supports a Many-to-Many relationship where a single account can be jointly owned by multiple family members (e.g., parents sharing a checking account).

- **Vendors and Normalization:** To ensure data consistency, the database maintains a list of Vendors (payees). Transactions are linked to these vendors to prevent spelling errors and allow for detailed reporting on where money is being spent.

- **Budgeting and Categorization:** To control spending, every transaction is assigned to a Category (e.g., Groceries, Rent, Entertainment). Each category has a defined MonthlyBudget limit, allowing the application to compare actual spending against the planned budget.

- **Financial Goals:** The database keeps track of long-term Budget Goals (e.g., "New Car Fund" or "Vacation"). It records the target amount, the deadline, and the current status of each goal, linked to the family member managing it.

- **Automation:** The system tracks Recurring Payments (such as monthly subscriptions or rent). These records store the amount and the specific day of the month the payment is due, allowing the application to auto-generate transactions or send alerts.

- **Security and Auditing:** For security purposes, the database includes an Audit Log. This creates a history of critical actions, specifically tracking when a transaction is deleted, recording the old data, the user responsible, and the timestamp of the deletion.

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• **Family Management:** The system supports multiple families. Each family is isolated with a unique FamilyID, ensuring that budget data and accounts are kept separate between different households.

2 ENTITY RELATIONAL MODEL

2.1 ENHANCED ER DIAGRAM

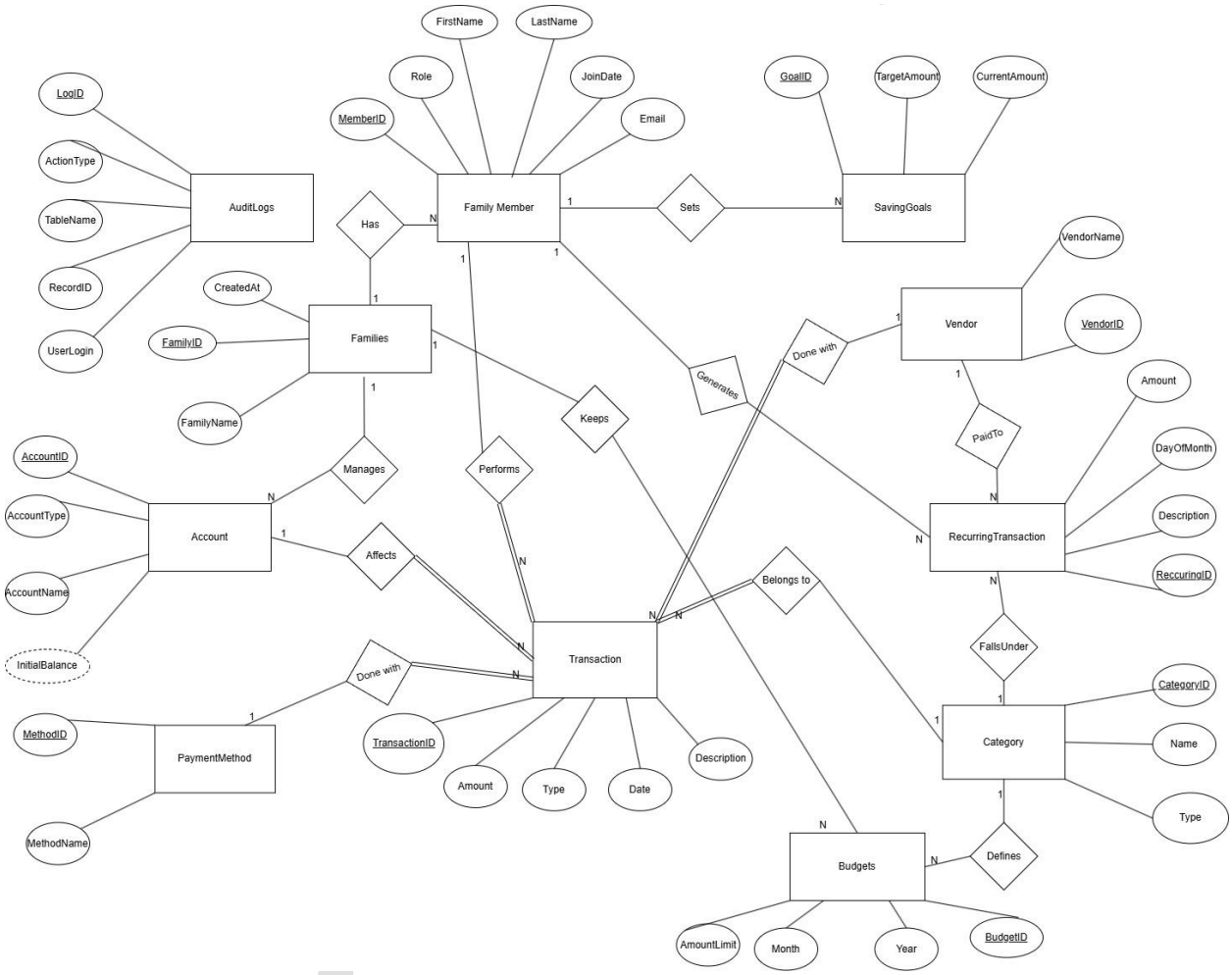

Figure 1: EER diagram of the Family Budget Database

2.2 RELATIONAL SCHEMA & MAPPING

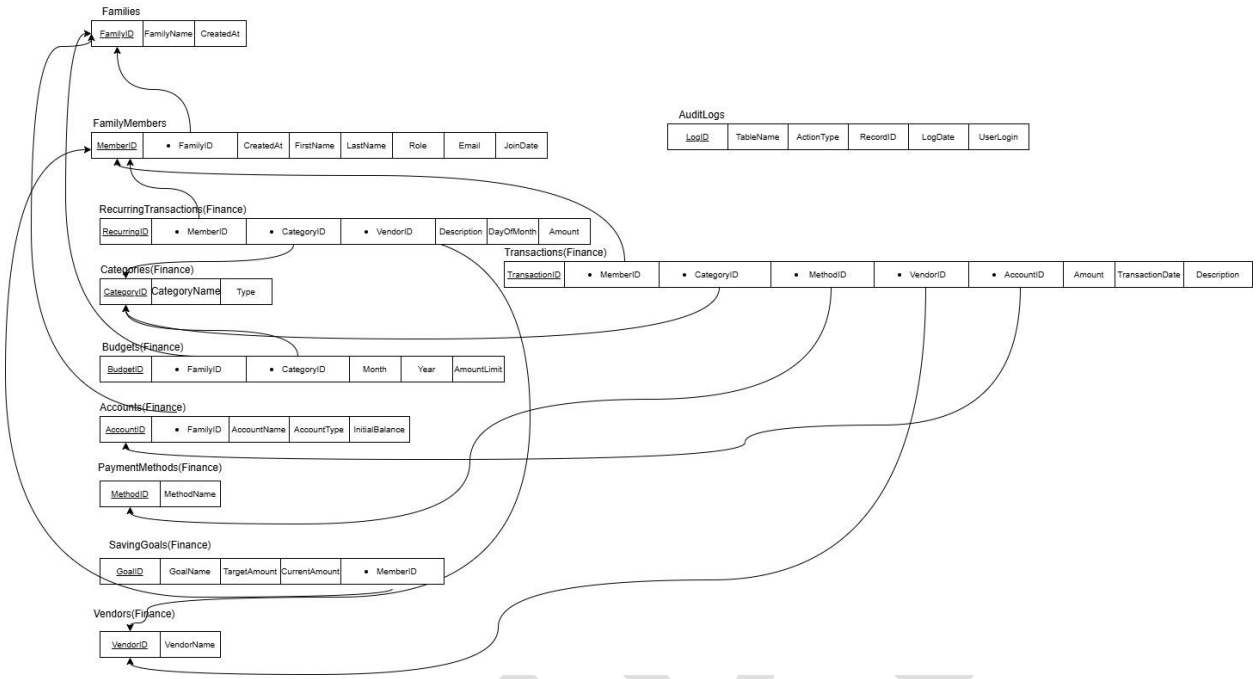

Figure 2: Relational schema of the Family Budget app Database with arrows indicating referential integrity

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3 NORMALIZATION

The normalization process was applied to the Family Budget Database to minimize redundancy and avoid insertion, update, and deletion anomalies.

3.1 FUNCTIONAL DEPENDENCIES

FD1 (FamilyMember): MemberID → FamilyID, FirstName, LastName, Role, Email, JoinDate

FD2 (Category): CategoryID → CategoryName, Type

FD3 (PaymentMethod): MethodID→ MethodName

FD4 (Vendor): VendorID → VendorName

FD5 (Account): AccountID → FamilyID, AccountName, AccountType, InitialBalance

FD6 (Budgets): BudgetID → FamilyID, CategoryID, Month, Year, AmountLimit

FD7 (SavingGoals): GoalID → GoalName, TargetAmount, CurrentAmount, MemberID

FD8 (Transactions): TransactionID → MemberID, CategoryID, MethodID, VendorID, AccountID, Amount, TransactionDate, Description

FD9(Families): FamilyID → FamilyName, CreatedAt

FD10 (RecurringTransactions): RecurringID → MemberID, CategoryID, VendorID, Amount, DayOfMonth, Description

FD11 (AuditLogs): LogID → TableName, ActionType, RecordID, LogDate, UserLogin

3.2 UNNORMALISED FORM

In the Unnormalized Form (UNF), we consider all the family budget data as a single, flat "universal spreadsheet" or report containing repeating groups.

Universal Relation: FamilyBudget_Report (TransactionID, TransactionDate, Amount, Description, MemberName, MemberRole, MemberEmail, AccountName, AccountType, AccountBalance, CategoryName, CategoryType, BudgetLimit, VendorName, PaymentMethodName, GoalName, GoalTarget)

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3.3 FIRST NORMAL FORM

To transform the relation into First Normal Form (1NF), we ensure that:

1. Each column contains only atomic (indivisible) values.
2. There are no repeating groups of columns.
3. Each record is unique.

We identify TransactionID as the primary key for the transaction data, but we still have a mixed schema.

Relation in 1NF: Family_Budget_1NF (TransactionID (PK), Date, Amount, Description, MemberID, FirstName, LastName, Role, CategoryID, CategoryName, BudgetLimit, AccountID, AccountName, InitialBalance, VendorID, VendorName, MethodID, MethodName)

3.4 SECOND NORMAL FORM

A relation is in Second Normal Form (2NF) if it is in 1NF and contains no partial dependencies. A partial dependency occurs when a non-prime attribute depends on only *part* of a composite primary key. In our design, most tables use a single surrogate Primary Key (e.g., TransactionID), which automatically satisfies 2NF conditions for those specific tables.

3.5 THIRD NORMAL FORM

In Transactions: TransactionID → MemberID → MemberEmail.

- The email depends on the member, not the transaction. We removed MemberEmail, Role, etc., from the Transaction table and placed them in the FamilyMember table

In Transactions: TransactionID → AccountID → AccountName.

- The account name depends on the account ID. We separated this into the Account table.

In Transactions: TransactionID → VendorID → VendorName.

- We separated vendor details into the Vendor table.

Final 3NF Schema**:** The database is now fully normalized into 3NF, resulting in the final list of tables:

FamilyMembers, Categories, PaymentMethods, Vendors, Accounts, Budgets, SavingGoals, Transactions, RecurringTransactions, AuditLogs,Families.

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